# Winding Is Not Enough: Involutive Compression and Exact Defect Modes in a Spinor-Derived Quantum Walk

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July 11, 2026 — draft v1 (overnight run)


### Abstract

In a discrete-time Dirac walk whose mass coin is derived from a local null-spinor field, the zeros of the field — collinearity points of the primitive spinors — are geometrically distinguished defects. We prove, machine-checked, an exact mode mechanism for displayed defect fixtures. First, the link data of any nowhere-zero field is *derived*: phase increments are principal arguments of neighbor ratios, their cycle total is an exact $2\pi$-integer winding, and a winding-one field is generated by explicit primitive spinors. Second, familiar candidate invariants are provably blind here: the walk determinant is $+1$ for every derived coin, every reflection-sector compressed determinant is $+1$ with or without walls, and every naive Lefschetz trace index vanishes identically — while exact $\pm 1$ modes nonetheless appear in a displayed two-wall walk. Third, its compression to the reflection-fixed sites is a self-adjoint unitary — an involutive compression, forcing exactly $(k \pm \operatorname{tr} M)/2$ modes at quasienergies 0 and $\pi$. Explicit rational left inverses prove that the complete displayed zero-wall and four-wall walks carry neither sign mode, and a second displayed complex-coin family has exact localized modes. Finally, an exhaustive classification of all 16 nowhere-zero sign fields on the four-cycle disproves the tempting winding-only explanation: two fields have the same derived winding $2\pi$ but different compression signatures, and the second compressed sector has neither sign mode. The exact repaired criterion is two walls together with a positional compatibility condition relative to the reflection-fixed sites, and it is now a machine-checked family-level law: the reflection-fixed-leg compression is self-adjoint precisely for the twelve minus four two-wall fields whose lone flip does not sit on a reflection-fixed site, and every such protected field's complete walk carries exact $+1$ and $-1$ eigenvectors through the compression engine (kernel-checked existence, one per sign). The exact multiplicities across all 16 complete walks — $\dim \ker(W \mp \mathbf{1}) = 2$ for each of the eight singleton fields, blind ones included, 4 for each of the four domain-block fields, and none for the zero- and four-wall fields — are now machine-checked kernel-only by explicit rank and kernel-vector certificates closed through rank–nullity (the retrofit removed every compiled-evaluator step); so the positional law delimits the reach of the reflection certificate, not mode existence. The standard refinements fare no better than the winding: the mirror-graded (reflection-sector) candidate is not merely blind but *ill-defined* on precisely the four blind fields, because the reflection commutes with the walk iff the two fixed sites carry equal signs, which every fixed singleton violates; and the two symmetric-time-frame windings, computed exactly on the bulk, are position-blind and equal on the counterexample pair. The counting engine and abstract theorems are kernel-checked in Lean 4; the positional-law core of the four-site family (`HalfPeriodInvariant`) is kernel-only after a 2026-07-12 retrofit, while the remaining four-site rational fixtures (mirror-chart, field-position, and full-walk control witnesses) use the compiled evaluator (documented), and the eight-site complex-coin witness is kernel-only. The classification layer of chiral-walk topology is imported, not claimed; the contribution is the derived origin of the defect data, the blindness pair, the involutive-compression engine, the exact fixture-level mode/no-mode separation, and the finite theorems that winding — and its time-frame and mirror refinements — underdetermine the compression while a value-only positional law determines it exactly. One further theorem and

one machine-checked impossibility complete the picture at this size. The certificate and its modes are stable across the *entire* coin continuum: for every real angle $\theta$, chart self-adjointness is *equivalent* to the vanishing of a single product — $M_{13}$ is self-adjoint iff $(\operatorname{sgn} b_0 + \operatorname{sgn} b_2)\sin\theta = 0$, and mirror-symmetrically for $M_{02}$ — so at every massive angle the symbolic criterion collapses to the positional one, every two-wall field's complete walk carries exact $\pm 1$ eigenvectors, and the wrong-chart failures are exactly $-2\sin\theta$; protection is an exact identity statement, not a perturbative one. And no translation-invariant index of the infinite periodic extension, composed with any decoder, reproduces the certificate: a protected singleton and its blind one-cell translate share the same periodic bulk. A general-length classification and a theorem deriving the reflection structure from the Pluecker carrier remain open.

# 1 Setup: derived defects, not assigned ones

The companion derivation paper [?] constructs the walk's complex mass coin from a local Pluecker field $z(x) = \psi(x) \wedge \phi(x)$ of primitive null spinors, with $z(x) = 0$ exactly at collinearity. Here the field enters only through its values. For any nowhere-zero field on a cycle, the link increment across an edge is the principal argument of the neighbor ratio, and:

**Theorem 1** (Derived winding)**.** *The cycle total of the derived increments is an exact integer multiple of $2\pi$, invariant under a global chiral phase; a globally lifted real phase forces zero; and an explicit winding-one field is the Pluecker coordinate of a primitive spinor pair field.*

Kernel (`PlueckerWindingDerived`.) The gauge content of the field is likewise classified: the complete orbit invariant of the walk under per-site chirality-diagonal gauges is the pair of chirality holonomies together with the local relative-phase current, and the walk is gauge-equivalent to its real-field form iff the phase is constant [?] (`GaugeClassification`, Kernel).

# 2 Every familiar invariant is blind

**Theorem 2** (Determinant blindness, both levels)**.** *For every value-derived rotation coin, wall count, and boundary condition,* $\det W = +1$*; and for the reflection-symmetric derived two-wall register, both reflection-sector compressed determinants equal* $+1$*, with or without walls.*

Kernel/Kernel+Eval (`signWalk_det_eq_one`; `sector_dets_all_one`.) Moreover $\operatorname{tr}\Gamma$, $\operatorname{tr}(\Gamma R)$, and the restrictions of $\Gamma$ to the $\pm 1$ eigenspaces all vanish identically across configurations, and the determinant-parity engine that would force a $-1$ mode from $\det = -1$ (`ChiralFlipMode`, Kernel) never fires, since every derived determinant is $+1$. Yet the two-wall walk has $\dim\ker(W \mp \mathbf{1}) = 2$ while the zero-wall control has none: something finer distinguishes the fixtures.

# 3 The involutive-compression engine and explicit fixture separation

**Theorem 3** (Involutive compression engine)**.** *Let $B$ be an isometry intertwining $W$ with a compression $M = B^\dagger W B$ that is self-adjoint and unitary. Then $M^2 = \mathbf{1}$, every characteristic root of $M$ is $\pm 1$ with multiplicities $(k \pm \operatorname{tr} M)/2$, and each root lifts through $B$ to an exact eigenvector of $W$ at quasienergy $0$ or $\pi$.*

Kernel (`InvolutiveCompression`, `involution_full_pinning`, `involution_trace_split`; no diagonalizability hypotheses.)

**Theorem 4** (Explicit defect/control separation)**.** *For the displayed reflection-symmetric walks on the four-site register, the compression of $W$ to the reflection-fixed sites is a self-adjoint unitary with* $\operatorname{tr} M = 0$ *in the two-wall configuration, forcing exact nonzero* $+1$ *and* $-1$ *modes. Explicit rational left inverses certify that the complete displayed zero-wall and four-wall walks have no nonzero eigenvector at either sign, with kernel-only polynomial-inverse certificates for the control annihilators (`PinnedControlAndBlind`,* Kernel*). Separately, an eight-site complex-coin fixture has exact localized mode witnesses, whose per-site geometric decay matches the exact* $\lambda = \pm 1$ *transfer matrices with eigenvalues* $\{\frac{1}{2}, 2\}$ *(`PinnedLocalization`,* Kernel*).*

Kernel/Kernel+Eval (`involutiveCompression_wall`, `twoWall_protected_modes`, `HalfWindingFullWalkCo` and `WallModeWitness`; the four-site rational fixtures use the compiled evaluator, documented, while the engine, counting, and eight-site witness are kernel-only.) The four-wall control is decisive methodologically: it lives in the same symmetry class — same grading, same reflection, determinant $+1$, genuine chirality — and has no modes, so no invariant of the symmetry algebra alone can express the separation.

## 4 The winding-only bridge fails, and the missing datum is positional

The three displayed zero/two/four-wall fixtures suggest a half-winding rule, but they do not test all fields with the same winding. The complete four-site family does.

**Theorem 5** (Complete four-site discriminant)**.** *For every nowhere-zero sign field on the four-cycle, the reflection-fixed-leg compression is self-adjoint, equivalently an involution, if and only if the field has exactly two walls and neither reflection-fixed site is isolated between two sites of the opposite sign. The compression is a nontrivial involution under exactly the same condition.*

Kernel+Eval (`HalfWindingFieldPositionClassification.discriminant`, `corrected_bridge`; exhaustive decision over 16 rational fields.)

**Theorem 6** (Same winding, different compression signature)**.** *The fields* $[+,+,-,+]$ *and* $[-,+,-,-]$ *both have derived total turning* $2\pi$*. The first compression is self-adjoint and supplies the displayed exact* $\pm 1$ *modes, while the second compression is not self-adjoint and its compressed sector has neither a* $+1$ *nor a* $-1$ *mode. Consequently no rule depending only on derived winding, wall count, or half-winding parity can determine the compression signature on this family.*

Kernel/Kernel+Eval (`winding_witness_eq_counterexample`, kernel-only; `winding_blind_to_signature`, `counterexample_sector_no_pos`, and `counterexample_sector_no_neg`, compiled finite evaluation.)

The positional condition is not read back from the compression. It is a field-level statement about whether site 1 or 3 is a sign singleton, so the repaired bridge is noncircular. It is also not topological: translating the same wall pattern relative to the fixed sites can change the signature while leaving its winding unchanged. Absence of $\pm 1$ modes for the second *complete* eight-dimensional walk is not proved here; the no-mode certificates in this theorem are compression-sector statements. In fact, exact rational linear algebra, machine-checked in `CensusMultiplicity` (kernel-only; integer kernel-vector certificates and invertible-minor rank certificates closed through rank–nullity; kernel-elaborated certificate products, no compiled evaluator), decides the complete walks of all 16 fields: *every* two-wall field carries both modes — each of the eight singleton fields, the four compression-blind fixed singletons included, has $\dim\ker(W \mp \mathbf{1}) = 2$, and each of the four domain-block fields has $\dim\ker(W \mp \mathbf{1}) = 4$ — while the zero- and four-wall fields have none. Mode existence at this size therefore follows wall count; what the positional law classifies exactly is the reach of the reflection-route *certificate*: the blind fields' modes exist but are not certified by the $\{1,3\}$-axis compression. The mechanism is now machine-checked: the certificate is *axis-equivariant*. Each two-wall field commutes with a site-axis reflection selected by its wall positions — the blind fixed singletons commute with the reflection through sites $\{0,2\}$, and the compression onto that axis's fixed legs is again a self-adjoint traceless unitary involution, with the mirror law $M_{02}(b)$ self-adjoint iff $b$ is two-wall and not a $\{0,2\}$-singleton (`PinnedMirrorChart.selfadj0_iff_mirrorProtected`, Kernel+Eval), so the same engine certifies their $2+2$ modes. The displayed law is thus one chart of a two-chart reflection atlas, and the 8-vs-4 split records which chart certifies a field. The domain blocks close the taxonomy in the sharpest way: their complete walks are themselves exact traceless self-adjoint involutions ($W^2 = \mathbf{1}$), so their $4+4$ modes are the engine applied at the identity compression. One engine, three compression levels, and the whole 16-field family is mechanistically explained — with the zero- and four-wall fields carrying no certificate and no modes. Two supporting layers are also machine-checked: the chart projectors are orthogonal spectral projectors into $\ker(W \mp \mathbf{1})$ of the certified rank (rank two per chart; rank four on blocks, where the count exhausts the space), by two kernel-only abstract lemmas whose chart instantiations consume the disclosed compiled-evaluator involution facts (`PinnedSpecProjectors`, Kernel/Kernel+Eval), and a third refinement kill joins the winding, timeframe, and mirror-winding blindness results: the symmetry-resolved chirality indices $\nu_{\varepsilon,r} = \mathrm{tr}(\Gamma P_r P_\varepsilon)$ vanish for *every* field with its applicable reflection — including the bond reflections on blocks — so chirality-type indices, global or sector-resolved, cannot separate protected from blind fields (`PinnedSectorIndex`, Kernel+Eval); the discriminator is the certificate boundary itself.

**Theorem 7** (Positional law and family-level protection)**.** *For every nowhere-zero sign field $b$ on the four-cycle: the reflection-fixed-leg compression $M(b)$ is self-adjoint if and only if $b$ has exactly two walls and is not a fixed singleton (a lone flip seated on reflection-fixed site 1 or 3); and for every such protected field the* complete *walk $W(b)$ carries an exact nonzero $+1$ eigenvector and an exact nonzero $-1$ eigenvector, produced by the involutive-compression engine. The four blind two-wall fields are exactly the fixed singletons.*

Kernel (`HalfPeriodInvariant.selfadj_iff_protected`, `protected_modes`; the engine composition and the 16-field decisions are now both kernel-elaborated — a 2026-07-12 retrofit replaced the module's compiled-evaluator decisions with kernel proofs, guard-pinned in `OvernightTheoryAxiomGuard` at footprint $\{$`propext`, `Classical.choice`, `Quot.sound`$\}$.)

**Theorem 8** (The mirror refinement is ill-defined exactly where needed)**.** *The reflection commutes with the derived walk iff the two reflection-fixed sites carry equal signs; every fixed singleton violates this. Hence the mirror-graded (reflection-sector) winding candidate is undefined on precisely the four blind fields it would need to separate.*

Kernel (`reflR_comm_walk_iff`, `fixedSingleton_not_reflSym`; both kernel-elaborated in the 2026-07-12 retrofit, footprint $\{$`propext`, `Classical.choice`, `Quot.sound`$\}$.) Separately, an exact rational computation (reported in the run record, not formalized) shows the two symmetric-time-frame bulk windings are $(0, \mp 2)$ per bulk sign — correctly predicting the domain-block modes — yet equal on the same-winding counterexample pair: time-frame refinements are position-blind, so they also cannot express Theorem **??**.

**Theorem 9** (Protection across the whole coin family)**.** *Parametrize the coin by a real angle: $W(b,\theta)$ has cosine $\cos\theta$ and sitewise signed sine $\pm\sin\theta$. For every real $\theta$: each domain block satisfies $W(b,\theta) = W(b,\theta)^\top$ and $W(b,\theta)^2 = \mathbf{1}$; whenever the $\{0,2\}$ signs cancel, the $\{1,3\}$-chart compression is a self-adjoint traceless involution intertwined with the walk, and mirror-symmetrically for the*

*$\{0,2\}$ chart whenever the $\{1,3\}$ signs cancel; consequently, for every two-wall field $b$ and every $\theta$, the complete walk $W(b,\theta)$ over $\mathbb{C}$ carries an exact nonzero $+1$ eigenvector and an exact nonzero $-1$ eigenvector. The identities hold identically in $\theta$ — every proof reduces to $\sin^2\theta+\cos^2\theta=1$ — so the certified modes persist across the entire mass family with no gap or continuity hypothesis. Conversely, the wrong-chart, zero-wall, and four-wall self-adjointness failures are the exact matrix entry $-2\sin\theta$: the classification is scoped precisely to the massive family $\sin\theta\neq 0$, and degenerates only at the massless points.*

Kernel (`ThetaFamilyProtection`: `modes_persist`, `block_involution_family`, `chart13_involution_family`, `chart02_involution_family`, `control_blind_not_selfadj`, `control_blind_massless`; and `ThetaFamilyComplet` `atlas_two_charts_family`, `M13_antisymm_entry`, `M13_selfadj_iff`, `M02_selfadj_iff` — the all-$\theta$ *iff*: chart self-adjointness is equivalent to $(\operatorname{sgn} b_0+\operatorname{sgn} b_2)\sin\theta=0$, every antisymmetric entry being that single monomial — and `Wth_eq_landed`, the transport pin proving the $\theta$-walk at $\cos\theta_0=\frac{4}{5}$, $\sin\theta_0=\frac{3}{5}$ equals the landed rational fixture walk, so the continuum statements provably contain the fixed-coin results. All kernel-only, no compiled evaluation; the sign-cancellation hypotheses are the natural generalization of the positional criterion.)

## 5 Relation to chiral-walk topology, and honest scope

Winding-number phases and $0/\pi$ boundary modes in chiral discrete-time walks are established physics [?, ?, ?, ?]; we import that classification and claim none of it. The two-time-frame structure of chiral walks and its half-period refinement are likewise established [?, ?], as are real-space symmetry indices for non-translation-invariant walks with their $\pm1$ bulk-boundary bound and stability theory [?, ?, ?]; our palindromic register walk is a symmetric-time-frame walk in exactly that sense, and the exact bulk time-frame windings on it are nontrivial. What is new here: (i) the defect data is *derived* from a local null-spinor field (Theorem ??), with defects at geometric collinearity points rather than assigned sign flips; (ii) a machine-checked blindness pair (Theorem ??) showing the standard discriminants cannot see the fixture-level separation; (iii) the involutive-compression engine (Theorem ??); (iv) the explicit fixture separation (Theorem ??); (v) the exhaustive four-site theorem showing that winding alone does not classify the compression and identifying the exact additional positional datum (Theorem ??); and (vi) the machine-checked positional law with family-level full-walk protection, together with the proof that the mirror refinement is ill-defined on exactly the blind fields (Theorems ??–??).

Boundaries. A winding-only half-index is refuted for the complete four-site sign census, and the time-frame and mirror refinements are refuted or ill-defined there as well; the value-only positional law is the proven replacement at this size. The relation to the real-space symmetry indices of Cedzich et al. [?, ?] is resolved at this size for the periodic-extension indices, via one explicit translation-orbit separation (and the standard translation-invariance of the CGGSVWZ indices, which we cite rather than formalize). On the finite register we claim no Fredholm invariant; in the Altland–Zirnbauer scheme the walk is class BDI, the constant bulks carry equal symmetric-frame windings (second-frame pair $\mp2$; transcribed from the exact oracle computation and flagged for source re-verification, not kernel-certified), the relative index across any single wall is bulk-determined, and the sector-resolved chirality indices vanish identically (`PinnedSectorIndex`, Kernel+Eval). The machine-checked impossibility (`CGGSVWZDictionary.no_periodic_index_reproduces_discriminator`, Kernel, plain `decide`) states that *no* translation-invariant index of the periodic extension, composed with any Boolean decoder, reproduces the certificate discriminator: a protected lone flip on a $\{0,2\}$-axis site and its blind one-cell translate onto a $\{1,3\}$-axis site are the same periodic bulk, so every such index identifies them while the certificate separates them. The discriminator is therefore strictly finer than any translation-invariant index of the periodic extension — and, by the standard fact that the CGGSVWZ indices are translation invariant, than every CGGSVWZ index — because it is anchored to a fixed reflection axis of the finite register, a datum translation-invariant bulk theory discards. (The Lean theorem quantifies over translation-invariant functions; no CGGSVWZ index object is itself constructed in the formalization.) General-$L$ classification and a theorem deriving the relevant reflection structure from the Pluecker carrier remain open. A second, complex-coin family on the eight-site ring has an exact nonzero $+1$ eigenvector and an exact nonzero $-1$ eigenvector over $\mathbb{Q}(i)$. For the displayed $+1$ mode, the proved site probabilities are $3277/64$ at sites 3 and 4 and 1 at sites 0 and 7; its zero-wall control has two-sided inverse certificates. Both families are integrated. The four-site instantiations carry the documented compiled-evaluator trust footprint; the eight-site witness is kernel-only. Nothing here predicts a mass value, and the walk families are 1+1-dimensional; higher-dimensional vortex defects ($z = f(r)e^{in\varphi}$) are the natural next construction.

# A Machine verification

Modules (all under `PhysicsSM/Draft/NullEdge/`): `PlueckerWindingDerived` (derived winding, Kernel); `HalfWindingFieldPositionClassification` (complete four-site discriminant and same-winding counterexample, Kernel/Kernel+Eval); `GaugeClassification` (orbit invariant, gauge triviality iff constant phase, Kernel); `ChiralFlipMode` (determinant-parity engine, Kernel); `SignWallDefectRouteB` and `SignWallDefectRouteBConcrete` (abstract sector engine and the determinant blindness pair, Kernel/Kernel+Eval); `ModeInvariantHalfWinding` (involutive-compression engine and displayed two-wall modes, Kernel/Kernel+Eval); `HalfPeriodInvariant` (positional law, family-level protection, mirror ill-definedness, chiral-frame structure, Kernel+Eval with kernel-only engine composition); `HalfWindingFullWalkControls` (complete zero/four no-mode controls, Kernel+Eval); `WallModeWitness` (second-family exact modes, localization comparison, and zero-wall control, Kernel); `PinnedMirrorChart` (the $\{0, 2\}$-chart mirror law and reflection structure, Kernel+Eval); `PinnedSectorIndex` (symmetry-resolved chirality indices vanish for every field, Kernel+Eval); `PinnedSpecProjectors` (spectral projectors into the certified eigenspaces: two kernel-only abstract lemmas plus compiled-evaluator chart instantiations, Kernel/Kernel+Eval); `PinnedLocalization` (exact $\lambda = \pm 1$ transfer matrices with eigendata, Kernel); `PinnedControlAndBlind` (exact polynomial inverses certifying the control annihilators, Kernel); `ThetaFamilyProtection` (whole-coin-continuum involution identities, modes, and exact controls, Kernel); `ThetaFamilyCompletion` (kernel all-$\theta$ atlas, the symbolic self-adjointness *iff* for both charts, and the landed-fixture transport pin, Kernel); `CGGSVWZDictionary` (the translation-invariance impossibility, plain kernel `decide`, with oracle-transcribed winding constants explicitly quarantined, Kernel); and `CensusMultiplicity` (the 16-field complete-walk multiplicity census 2/4/0 for singleton/domain-block/control fields, by integer kernel-vector and invertible-minor certificates closed through rank–nullity, Kernel+Eval). Assumption footprints are build-enforced in the repository's aggregate guard module (`OvernightTheoryAxiomGuard`), which pins each named headline theorem's exact axiom list — the standard three axioms for the kernel-marked results, plus the documented compiled-evaluator pair on the explicit rational instantiations only — and fails the build on any drift; `PlueckerPairGenerator`-style in-file guards supplement this where noted.

# References


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